

Problem 1:

The endothermic liquid phase elementary reaction $A+B \rightarrow 2C$ proceeds, substantially, to completion in a single steam jacketed, CSTR. Calculate the steady-state reactor temperature.

Reactor volume: 125 gal

Steam jacket area: 10 ft²

Jacket steam: 150 psig (365.9°F saturation temperature)

Overall heat transfer coefficient of jacket, $U=150$ Btu/h.ft².F

Agitator shaft horsepower: 25hp

$\Delta H_{Rx}^0 = +20,000$ Btu per lb mol of A (Independent of temperature)

	A	B	C
Feed (lbmol/hr)	10	10	0
Feed temperature (F)	80	80	
Specific heat (Btu/lbmol . F)	51	44	47.5
Molecular weight	128	94	222
Density (lb/ft ³)	63	67.2	65

Problem 2:

The irreversible liquid-phase reactions (1) $A+B \rightarrow 2C$. $r_{1C} = k_{1C}C_A C_B$; and (2) $2B+C \rightarrow D$. $r_{2D} = k_{2D}C_C C_B$

Are carried out in a PFR with heat exchanger. The concentration of A, B, C and D were measure at the point down the reactor where the liquid temperature of the reactor reach a maximum at 500 K and coolant stream is at 325 K. They were found to be $C_A=0.1$; $C_B=0.2$; $C_C=0.5$; $C_D=1.5$ all in mol/dm³. The product of the overall heat-transfer coefficient and the heat exchanger area per unit volume is 10cal/s.dm³.K. The entering molar flow rate of A is 10 mol/s. What is the activation energy of reaction (1) in cal/mol?

Information:

$$C_{P_A} = C_{P_B} = C_{P_C} = 30 \text{ cal/mol.K}; C_{P_D} = 90 \text{ cal/mol.K}$$

$$\Delta H_{Rx1A} = -50,000 \text{ cal/molA}$$

$$\Delta H_{Rx1A} = +5000 \text{ cal/molB}$$

$$k_{1C} = 0.043 \frac{\text{dm}^3}{\text{mol.s}} \text{ at } 400\text{K}$$

$$k_{2D} = 0.4 \frac{\text{dm}^3}{\text{mol.s}} e^{5000 \text{ K} \left[\frac{1}{500} - \frac{1}{T} \right]}$$